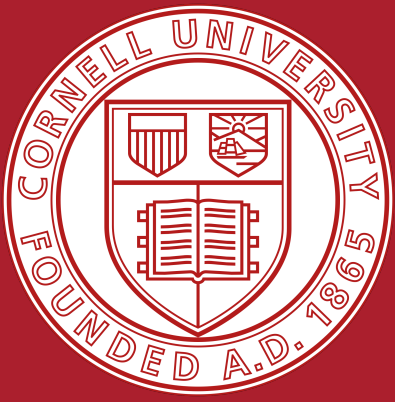




LoRA++: Variations on Low-rank approximations for GPT-2 fine-tuning

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Motivation

Fine-tuning normally means keeping a new full model for every task. For large language models, this becomes expensive fast.

LoRA's idea

Instead of retraining an entire matrix W , we train low-rank approximations. We do so because we believe that the features necessary for a fine-tuned task live in a much lower dimension.

LoRA Architecture

frozen pretrained weight W_0

$A: r \times n$

$B: n \times r$

Train only A, B (rank $r \ll n$). Forward pass: $h = W_0x + \frac{\alpha}{r}BAx$.

Experiment specification

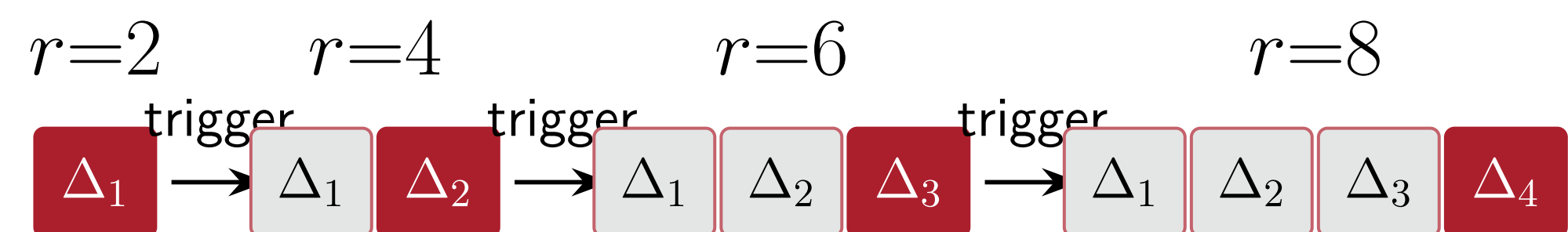
- Original target: LoRA Table 3, GPT-2 on E2E NLG.
- E2E NLG: restaurant facts $\rightarrow$ fluent description.
- Our scale: GPT-2 Small, full fine-tune vs. LoRA $r = 2, 4, 8$.
- Metrics: BLEU and ROUGE-L, both text-overlap scores.
- Implementation: LoRA written from scratch, no PEFT.

Setup at a Glance

Model	GPT-2 Small, 124.4M params
Task	E2E restaurant text generation
LoRA ranks	$r = 2, 4, 8$
LoRA trainable	73.7K–294.9K params
Biggest saving	1,688 $\times$ fewer trainable params

Sequential LoRA (SLoRA)

We investigated the effect of adding ranks incrementally-instead of starting with a rank 8 approximation, we start at rank 2 and only add more ranks if our loss curve demands it. In other words, we perform LoRA iteratively, adding ranks in a sequential manner until convergence.



Trigger fixed every N steps or validation-loss plateau
 α on 0 frozen ($\equiv$ Chain-of-LoRA [4]) $\cdot$ 0.1 hybrid $\cdot$ 1.0 unfrozen

Experiment Adaptations

Result	Trainable	BLEU	ROUGE-L
Paper: GPT-2 Medium full FT	354.9M	68.2	71.0
Paper: GPT-2 Medium LoRA	0.35M	70.4	71.8
Ours: GPT-2 Small full FT	124.4M	64.76	68.24
Ours: GPT-2 Small LoRA $r = 8$	294.9K	65.21	67.23

The paper says LoRA can match full fine-tuning with far fewer trainable weights. We test that claim at a smaller compute budget by using GPT-2 small.

LoRA Benchmarking Results

Experiment	Trainable	% Total	BLEU	ROUGE-L
Full fine-tune	124.44M	100.000	64.76	68.24
LoRA $r = 2$	73.7K	0.059	64.20	67.39
LoRA $r = 4$	147.5K	0.118	64.34	67.87
LoRA $r = 8$	294.9K	0.236	65.21	67.23

LoRA kept quality close while training 400x–1,688x fewer parameters.

LoRA stays close to fine-tuning

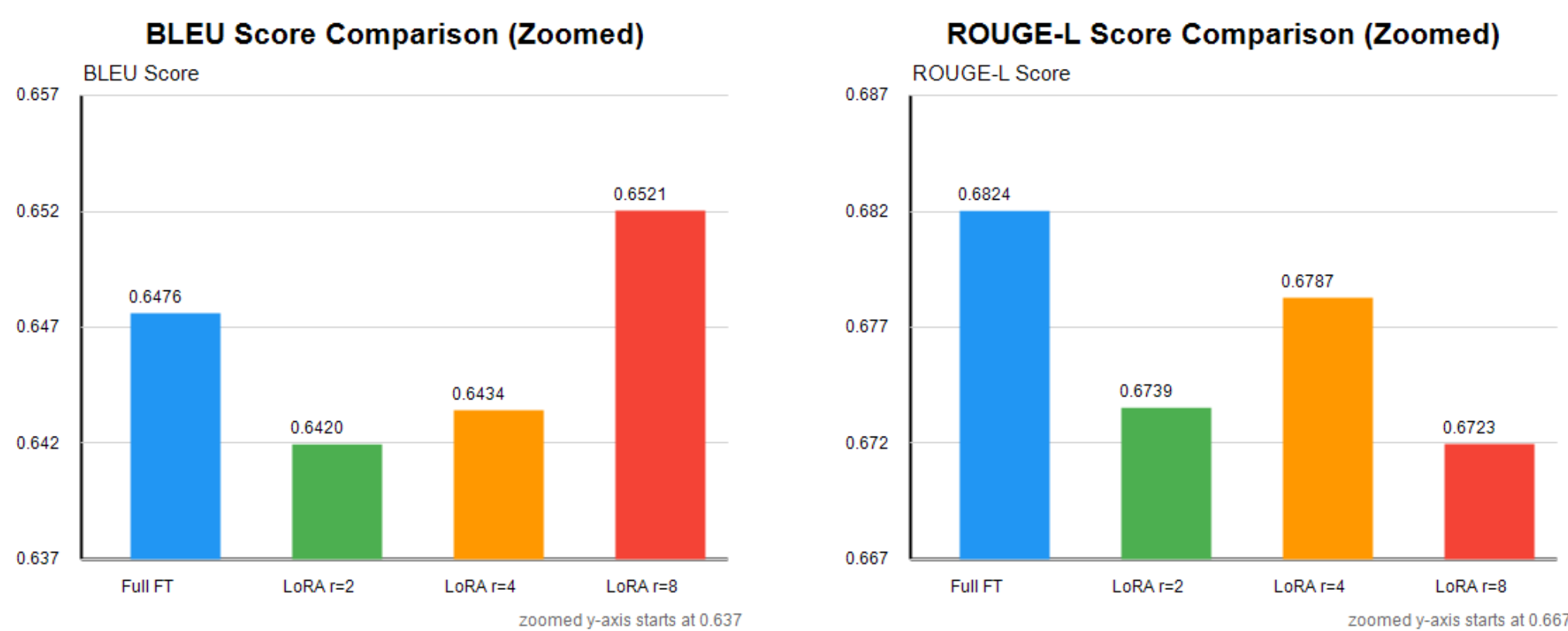
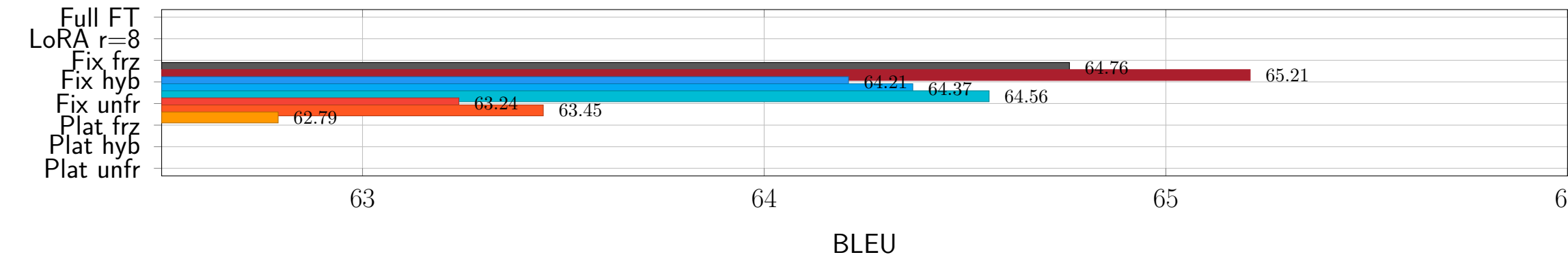


Figure 1. BLEU and ROUGE-L for full fine-tuning and LoRA ranks.

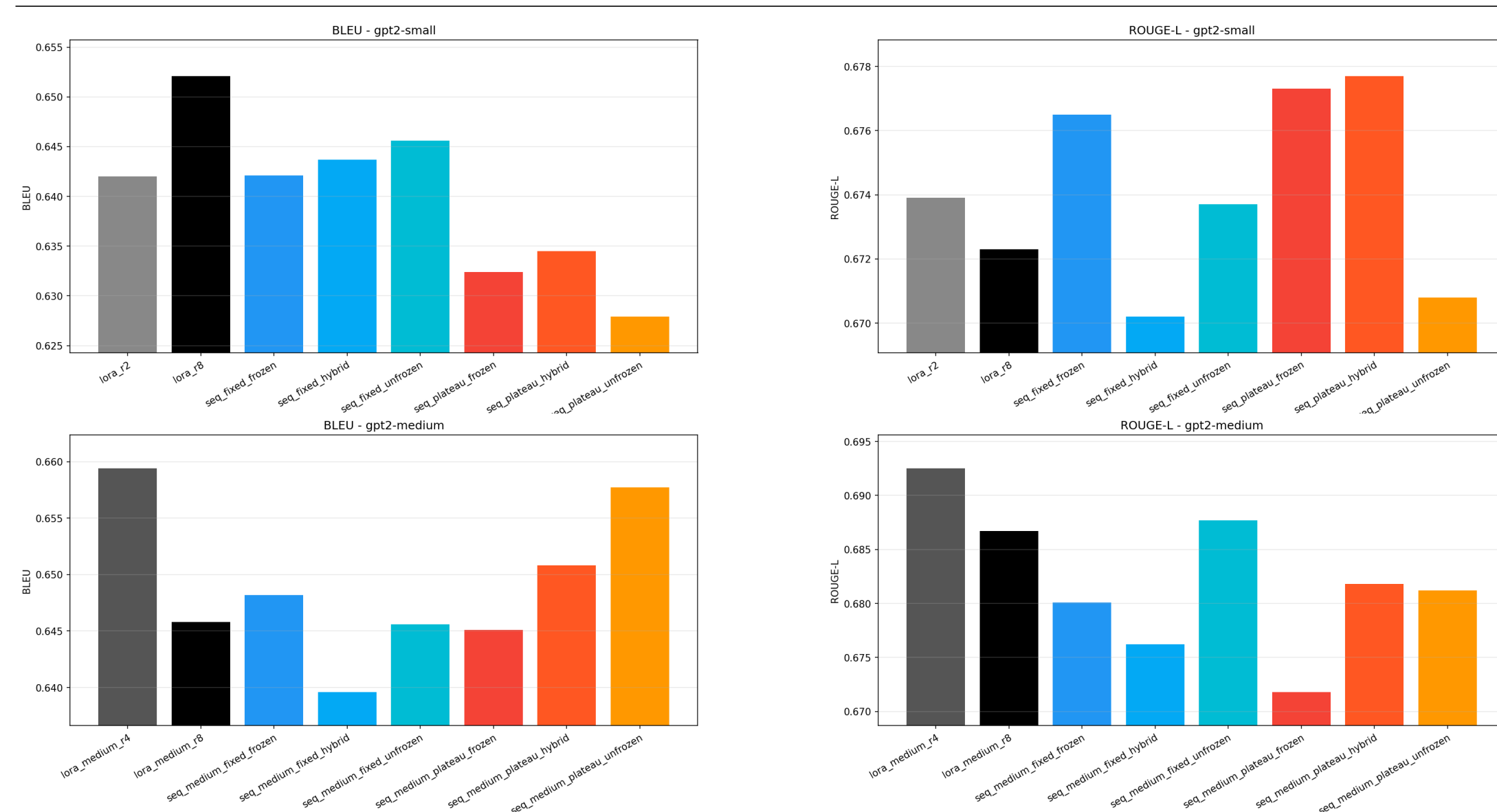
References

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SLoRA BLEU on GPT-2 Small

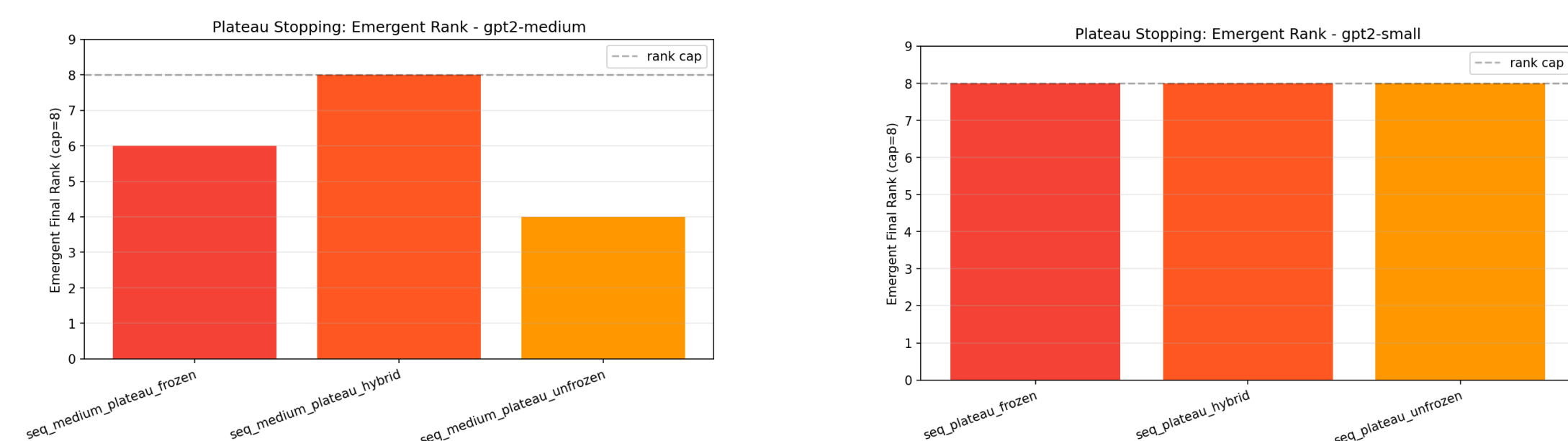


SLoRA results



Gray = LoRA $r=2$ (small) / $r=4$ (medium); black = LoRA $r=8$. Blues = fixed-step trigger, reds/oranges = plateau trigger.

Stopping ranks for SLoRA



Where the plateau trigger settled. Medium found a real stopping point (**rank 4** at $\alpha=1$, rank 6 frozen, hybrid at the cap). Small lacked this behavior, as all three runs saturated at 8, suggesting that there may be additional dimensionality constraints for smaller models.

Conclusion

On GPT-2 Medium, plateau trigger with $\alpha=1$ rediscovered **rank 4** on its own and beat fixed rank-8 LoRA on BLEU (**65.77** vs. 64.58). Adaptive rank discovery worked without sweeping r . While sequential LoRA doesn't significantly outperform standard LoRA techniques, we find the ability to automatically discover the rank of the result, an important hyperparameter, to still hold significant value in reducing the time needed to find an optimal rank.